# From the Atari Shock to a Modern Crisis: Analyzing Mass Layoffs in the Post-Pandemic U.S. Video Game Industry

JIESI MA, Washington University in St Louis, St. Louis, United States

Since 2023, the video game industry (VGI) has faced a severe workforce reduction marked by widespread layoffs, studio closures, and project cancellations, with the United States being the most affected nation. This study employs comparative analysis to investigate the underlying drivers of this layoff crisis, drawing parallels with the 1983 Video Game Crash (Atari Shock) to identify historical commonalities and contemporary divergences. Using a three-level framework, it analyzes macroeconomic shifts (e.g., monetary policy changes), industry dynamics (e.g., management inefficiencies, rising costs), and project-level risks (e.g., failed live-service transitions, imitative design) to interpret systemic vulnerabilities through established theories in business studies, project management, and game development. Notably, the industry's resilience—rooted in globalized production networks and diversified revenue streams—has averted systemic collapse, instead containing the crisis as a supply-side turbulence.

CCS Concepts: • **Social and professional topics → Employment issues**;

Additional Key Words and Phrases: Video game industry, video game history, layoffs, video game development, monetary policies, corporate management, project management

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## 1 Introduction

After decades of development, the **video game industry** (**VGI**) has become a global industry [1], providing hundreds of thousands of direct and indirect job opportunities worldwide. However, since late 2022, VGI has faced major workforce cuts, with mass layoffs and studio closures posing profound challenges [2]. During this period, widespread project cancellations and organizational downsizing led to 10,500 job losses in 2023, 14,600 in 2024, and a continuing trend in 2025 [3]. Notably, every week of 2024 saw layoffs or business closures, disproportionately impacting North America and Europe—with the U.S. most affected [4]. The **Game Developer Conference** (**GDC**) 2025 report notes 11% of developers experienced layoffs in the prior year. Corporations cite restructuring, declining revenue, and market shifts as reasons, while developers attribute layoffs to COVID-19 effects, corporate greed, and previous over-hiring [5].

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Author's Contact Information: Jiesi Ma, Washington University in St Louis, St. Louis, Missouri, United States; e-mail: arcboundavenger@gmail.com.



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The VGI's ongoing employment downturn is causing widespread workforce reductions and project cancellations, forcing many companies to reorganize. However, current literature lacks comprehensive analysis of this event, highlighting this study's necessity. This study focuses primarily on the impacted U.S. VGI—specifically examining publicly traded entities and their subsidiaries, alongside startups and mid-sized companies primarily backed by **private equity** (**PE**) or **venture capital** (**VC**) funding, which have disclosed the most extensive layoff data. This segment includes global giants like Microsoft Gaming, Sony Interactive Entertainment, and Electronic Arts, plus most U.S.-based mid-sized video game companies. While small and independent studios now employ comparable numbers to large firms [8], mid-to-large sized video game companies are still widely recognized as the essential part of VGI that drives industry practices [9]. Moreover, limited public data on workforce reductions at independent teams places this segment outside the core scope of this study.

As the first systematic investigation of mass layoffs in the U.S. VGI, this study seeks to examine the underlying drivers of the ongoing industry crisis. By conducting a comparative analysis with the Video Game Crash of 1983—a major collapse event in gaming history—this study aims to illuminate both historical parallels and contemporary distinctiveness in their underlying mechanisms. The findings offer a historically grounded understanding of recent mass layoffs and provide actionable strategies for industry stakeholders to navigate the current crisis challenges effectively. This dual-temporal approach contributes meaningful perspectives to academic discourse on VGI dynamics while supporting long-term industry sustainability.

## 2 Research Questions

This study investigates the following interconnected questions, examining the phenomenon of massive layoffs within the U.S. VGI through both historical and contemporary lenses:

- What potential factors contributed to the mass layoffs across U.S. VGI following the COVID-19 pandemic?
- How do the mechanisms driving post-pandemic workforce reductions compare to those during the Video Game Crash of 1983?
- What actionable strategies can developers employ to enhance resilience?

By addressing these questions, this research aims to provide a holistic and historically grounded understanding of the recent waves of layoffs in the U.S. VGI. The findings seek to yield insights pertinent to the present industry context while contributing meaningful perspectives to the academic discourse on VGI dynamics.

## 3 Background and Context

It is well known that the video games originated in the U.S. [10]. The origin of U.S. VGI was intrinsically linked to the 1950s–1960s nascent U.S. computer industry, drawing talent from early computer hardware ecosystem around **Massachusetts Institute of Technology** (**MIT**) and Silicon Valley [11]. MIT graduate Steve Russell programmed one of the earliest games, *Spacewar!*, on a PDP-1 computer in 1962 [12]. In 1972, Nolan Bushnell founded Atari, which earned traction with *Pong* [13]. In 1977, Atari achieved dominance in the U.S. home console market with the success of its Atari 2600 [14].

Until 1979, Atari retained exclusive software development rights for its Atari 2600. Although maintained dominance in the late 1970s, Atari released numerous overhyped yet poorly designed games, such as *Superman* [15], due to mismanagement. Then, four engineers upset at their treatment by Atari left to form Activision. Atari initially threatened legal action but eventually adopted a royalty system for third-party developers. This shift led to a sharp rise in third-party

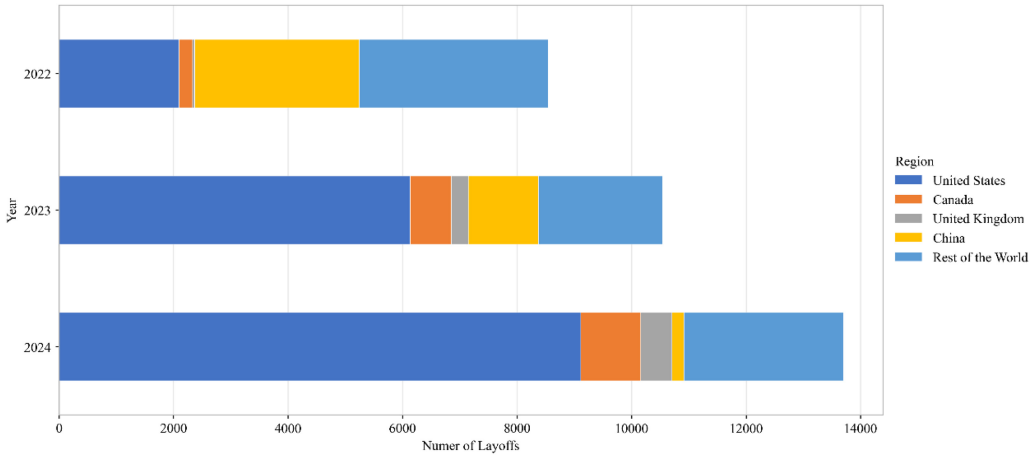



Fig. 1. Gaming industry layoffs by country, 2022–2024. *Source*: Adapted from Gaming Industry Layoffs tracker [6]. *Note*: Layoffs in China in 2022 were driven by a temporary pause in game licensing approvals, followed by a rapid recovery as approvals resumed [7].

game developers, fueled by VC [16]. By 1982, the U.S. VGI had expanded into a multi-billion-dollar market, with over 150 third-party developers having released games for the Atari 2600 [16]. Lowered barriers enabled even non-gaming entrants like Quaker Oats to release many poorly received titles [17]. Fierce competition and talent shortages caused rampant game replication. Atari 2600 ROM analysis shows declining lexical diversity and information complexity amid increasing releases [18]. By 1983, imitators outnumbered expert developers 4:1, leaving most games homogeneous. The flood of inexperienced and underfunded developers caused a glut in the video game market [19]. However, due to their small size, lack of knowledge, limited resources, and weak market presence, these new entrants were particularly vulnerable [20]. Eventually, combined with a market flooded with low-quality game clones and tightening monetary policies in the U.S. [21], the commercial failure of *E.T. the Extra-Terrestrial* triggered the Video Game Crash of 1983 [22], an industry-wide collapse that plunged the U.S. VGI into severe recession.

The Video Game Crash of 1983, also known as the Atari Shock, is a major industry collapse which saw U.S. home gaming revenues plummet from $3.2 billion to $100 million between 1983 and 1985 (see Figure 2) [24]. This crisis driving most U.S. video game companies out of business while personal computers (PCs) emerged as disruptive innovators concurrently. As consumer confidence in consoles waned, users shifted to PCs for entertainment, thereby eroding the home console market share [25], worsening the downturn. Fortunately, constrained by regionally fragmented video game markets in 1980s, this crisis did not reach Japan. However, the Atari Shock still caused long-term revenue declines in U.S. and global video game markets for over a decade (see Figure 3).

The U.S. video game market's decline finally reversed with the launch of **Nintendo Entertainment System** (**NES**) in U.S. in 1985 [28], which initiated the U.S. VGI's gradual recovery through rigorous quality control. After the Atari Shock, many developers shifted from console to PC game development, with 1982-founded Electronic Arts among the influential companies established in this period. As PCs gained widespread adoption in the 1980s, two parallel gaming markets emerged in the U.S.: PC gaming and console gaming [29].

In the 1990s, hardware and software advances drove rapid VGI transformation. Key milestones, such as Microsoft's Windows 95/DirectX (1995) and NVIDIA's GeForce 256 (1999) redefined the PC gaming infrastructure. Microsoft's 2001 Xbox launch subsequently established it as a major

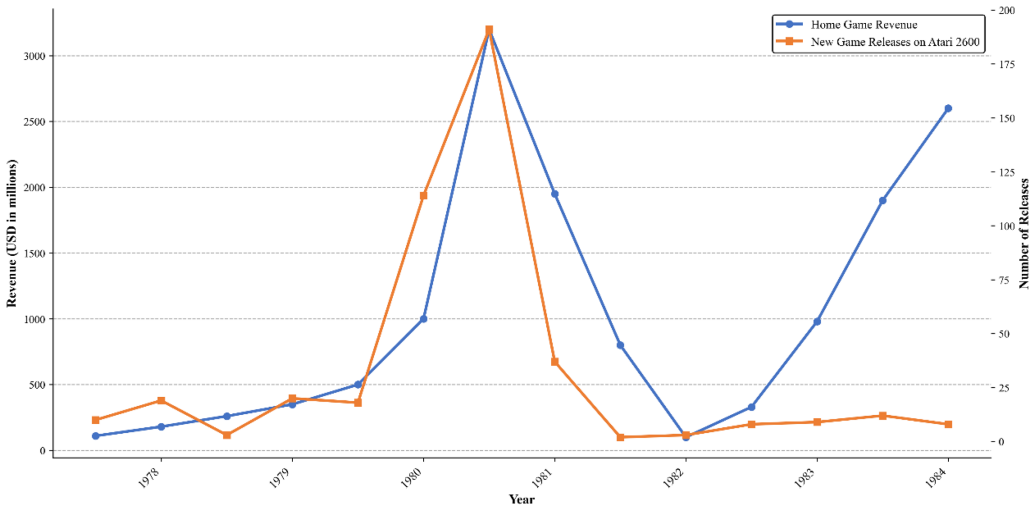



Fig. 2. U.S. home game annual revenue (millions of dollars, inflation adjusted to 1983) and number of new game releases on Atari 2600. *Source*: Adapted from Ernkvist [16] and MobyGames [23]. *Note*: While the accuracy of specific data points may be subject to uncertainty, the overall trend is likely representative.

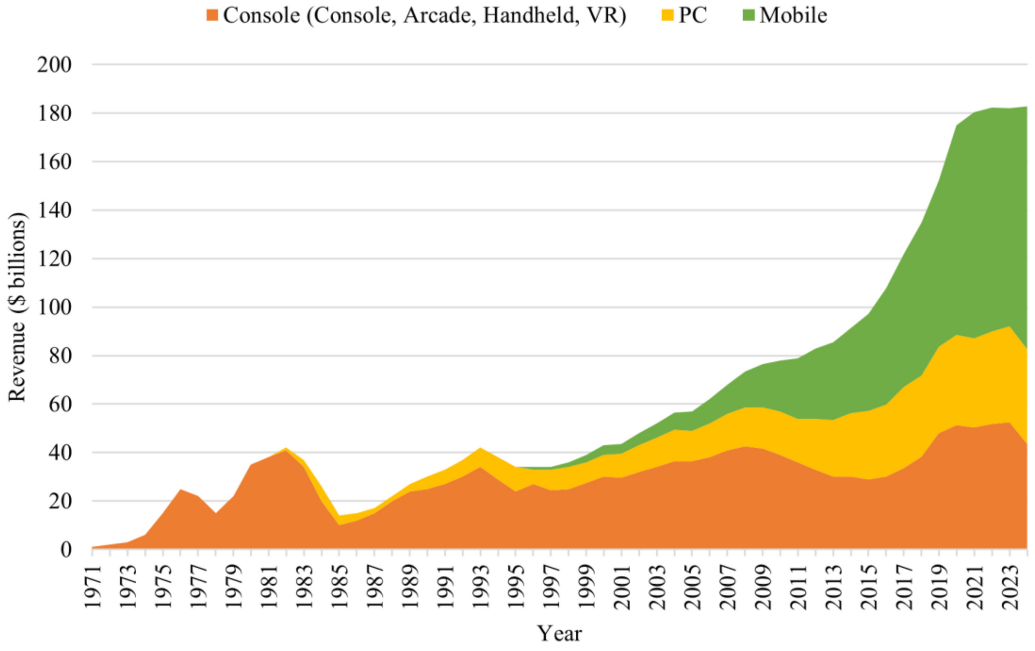



Fig. 3. Global VGI revenue from 1970 to 2024. *Source*: Adapted from Bloomberg [26] and Newzoo [27].

console player yet around the same time, PC gaming began declining relative to consoles due to piracy enabled by expanded Internet access. However, the PC market downturn was reversed by Valve's digital distribution platform Steam [30], which fundamentally transformed the business and distribution model of PC games: digital sales accounted for less than 50% of PC game sales in 2010 but rose to over 90% by 2013 [31].

The 2007 iPhone launch was a major turning point, greatly expanding global Internet access and the player base via mobile. Capitalizing on this reach and the **free-to-play** (**F2P**) model's low barrier to entry, smartphones grew to dominate global gaming revenue over the 2010s [32]. Simultaneously, globalization of VGI accelerated, enabling countries like Poland, China, and South Korea to narrow the gap in game development with traditional hubs like the U.S. and Japan. When it comes to mobile gaming, while companies like Zynga and Scopely have achieved success, no U.S. mobile game developer achieved dominance in the mobile sector comparable to the strong position U.S. companies hold in PC and console gaming.

## 4 Methodology

Over the past decades, scholars have proposed different explanations for the Atari Shock. A study by Ernkvist proposes a key framework called the "3Ds" model [16]. This analytical framework attributes the systemic failure to: (1) *Declining entry barriers* that caused market saturation; (2) *Decreasing product differentiation* in the market; (3) *Disruptive innovations* that obsoleted incumbent firms' core competencies. This model explains how these factors combined to bring about the Video Game Crash of 1983 and will also be adopted in this study.

Through comparative analysis [33], this study investigates the causes of the current layoff crisis by comparing it with the historical Atari Shock. Drawing on media sources, published histories of early VGI, and academic analyses, it centers on the business aspects of VGI. Additionally, while the "3Ds" framework explains the Atari Shock's primary causes, it overlooks the economic context in which early 1980s game projects operated at a significantly smaller scale. Therefore, analyzing modern phenomena like post-COVID layoffs requires expanding the "3Ds" framework through economic, project management and game development lenses. This study analyzes the current crisis using a three-level framework: Macro (economic conditions), Meso (industry dynamics), and Micro (game development practices). Key factors within each level are interpreted using these disciplinary perspectives. The discussion compares factors by consolidating findings from the current crisis analysis with literature-defined 1983 crash factors, identifying critical convergences and divergences.

## 5 Macro-Level: Economic Conditions

In the 2010s, the VGI underwent a decade of rapid growth. This led industry practitioners and observers to become accustomed to the growth, frequently overlooking the impact of macroeconomic factors on the sector. Yet history provides warnings: As Figure 3 indicates, PC/console sectors endured a three-year downturn following the 2008 Great Recession [32]. Going further back, an often-overlooked fact is that the Atari Shock was also associated with the global recession of 1982 [21]. These historical contractions clearly reveal the VGI's vulnerability to macroeconomic trends. Similarly, the VGI's growth during the COVID pandemic stemmed from converging macroeconomic forces—including expansive monetary policies and surging consumer demand—yet this growth trajectory reversed quickly after the macroeconomic conditions shifted. Analysis of this boom-and-bust cycle from 2020 to 2025 contextualizes recent mass layoffs and reveals patterns that may help mitigate future crises.

### 5.1 The Impact of Monetary Policies

The U.S., home to the world's most developed capital market [34], sees industries shaped by its capital flows and ripple effects. Like other sectors, the VGI was deeply affected by COVID-19 and the concurrent monetary policies, with major advanced-economy central banks implementing **quantitative easing** (**QE**) from 2020–2022 to counter the pandemic's economic disruptions [35]. This expansionary policy empirically demonstrated to accelerate capital deployment in investments

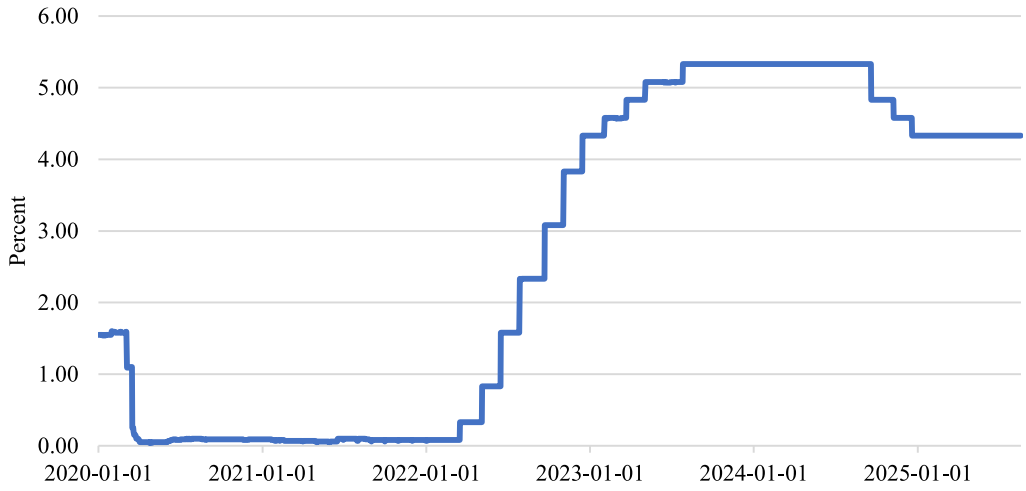



Fig. 4. Federal funds effective rate, 2020–2025. *Source*: adapted from federal reserve economic data [49].

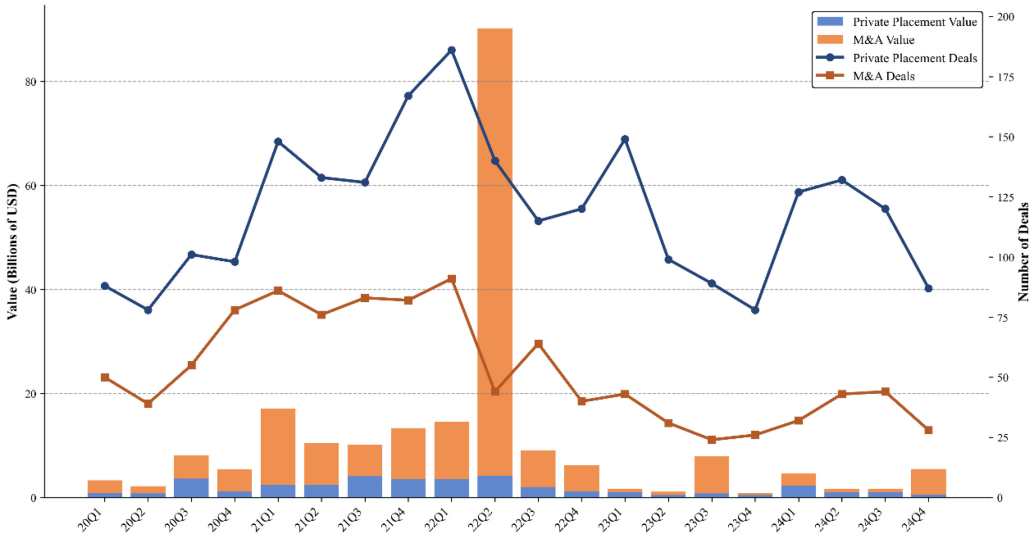



Fig. 5. Gaming investment value and deals count, 2020–2024. *Source*: Adapted from InvestGame's Global Gaming Deals Report [50].

[36]. Accordingly, publicly traded gaming corporations saw huge share price appreciation throughout 2020. Lockdowns made gaming a primary global entertainment source concurrently [37]. During this period, Steam's peak concurrent online user count increased from 18.33 million in January 2020 to 29.20 million in January 2022 [38]. Newzoo reports that VGI revenue grew by 24.7% year-on-year, from $139.0 billion in 2019 to $173.4 billion in 2020 [27]. These demand-side dynamics interacted with QE-induced liquidity, creating a synergistic market environment.

As shown in Figure 4, QE implemented by the Federal Reserve lowered interest rates to near-zero levels [39], significantly reducing financial costs for corporations [40]. This easing monetary environment enabled large firms to engage in extensive **merger and acquisition** (**M&A**) activity, as well as minority investments, within the VGI (see Figure 5). With the largest capital market,

the U.S. experienced a more intense investment surge in VGI compared with other nations. In 2021, Microsoft acquired Zenimax for $7.5 billion [41], Electronic Arts purchased Glu for $2.4 billion [42], and Embracer Group bought Gearbox Software for $1.3 billion [43]. The momentum intensified in 2022, when Take-Two Interactive set a record for the largest acquisition in the VGI by purchasing Zynga for $12.7 billion [44]. However, this record was broken less than two weeks later by Microsoft's monumental $75.4 billion acquisition of Activision Blizzard [45]. Sony followed suit with its $3.7 billion acquisition of Bungie [46]. Statistically, 13 of the 20 largest M&A deals in gaming history occurred between 2020 and 2022. The value of all investments in the VGI per quarter, including private placements and M&A, surged from approximately $2 billion in Q1 2020 to about $10 billion in Q4 2021 [47]. Meanwhile, quarterly deals grew from under 100 to nearly 300. By Q1 2022, the total investment amount spiked to nearly $100 billion due to Microsoft's acquisition of Activision Blizzard, and the deal count also increased to nearly 400 [48], both setting historical records. This period of consolidation fundamentally reshaped the VGI in the U.S. as major corporations expanded their dominance through M&A.

Although VC and PE are generally not the preferred financing source for game developers—previous research indicates game developers tend to favor publishers or outsourcing work for funding [51]—the U.S. VGI presents an exception. Due to its deep historical ties to the tech industry and the well-developed VC ecosystem, startup companies in the U.S. VGI more used to seeking through equity investments from VCs [52]. As shown in Figure 5, this rise in liquidity and historically low interest rates also facilitated VC investments and led to the overvaluation of startup companies, catalyzing the formation of bubbles [53]. Notably, despite performing well during financial downturns, PE tends to underperform when both the stock market and enterprise values are already high due to cyclicality and liquidity risks [54].

This financial environment led to widespread over-hiring practices not only in the technology industry but also in the gaming sectors. Federal Reserve Economic Data reveals that software development job postings on Indeed in the U.S. doubled from their early-2020 levels by mid-2022 (see Figure 6) [55]. To counter productivity losses from remote work [56], capital-flush major corporations deployed aggressive acquisitions and workforce expansions, while VC-backed startups also competed for talent. Consequently, over-hiring in U.S. VGI became pronounced.

The VGI from 2020 to 2022 displayed a thriving landscape, with industry seemingly more optimistic about the future of gaming than ever before. S&P once estimated that the global gaming market's revenue could grow to $200 billion in 2022 [57]. However, the growth trajectory reversed sharply starting in 2022, coinciding with the Federal Reserve's initiation of interest rate increasement [58]. These rate hikes triggered a sharp contraction in market liquidity and increased financing costs across the economy [59], putting immense pressure on game developers. By the first quarter of 2023, quarterly investment inflows into the VGI sector had declined precipitously, marking a dramatic contraction compared with the preceding boom [60].

### 5.2 Stagnation of Player Growth

The mobile Internet boom of the 2010s triggered player population expansion, a trend amplified by COVID-19 lockdowns. This expansion resulted in global players increasing from 1.99 billion in 2015 to 3.42 billion in 2024 [61], yet the growth rate declined from over 10% in 2021 to under 2% in 2024 – reaching their lowest level in a decade (see Figure 7). Furthermore, most player growth came from emerging regions with limited purchasing power [62].

Concurrently, liquidity contraction stemming from tight monetary policy also impacted the demand side. In 2024, global gaming revenue experienced an annualized growth of only 0.5%, with North America—the region most severely affected by layoffs—recording a minimal annualized growth rate of 0.1% [27]. This sustained low-growth trend not only reflects the short-term

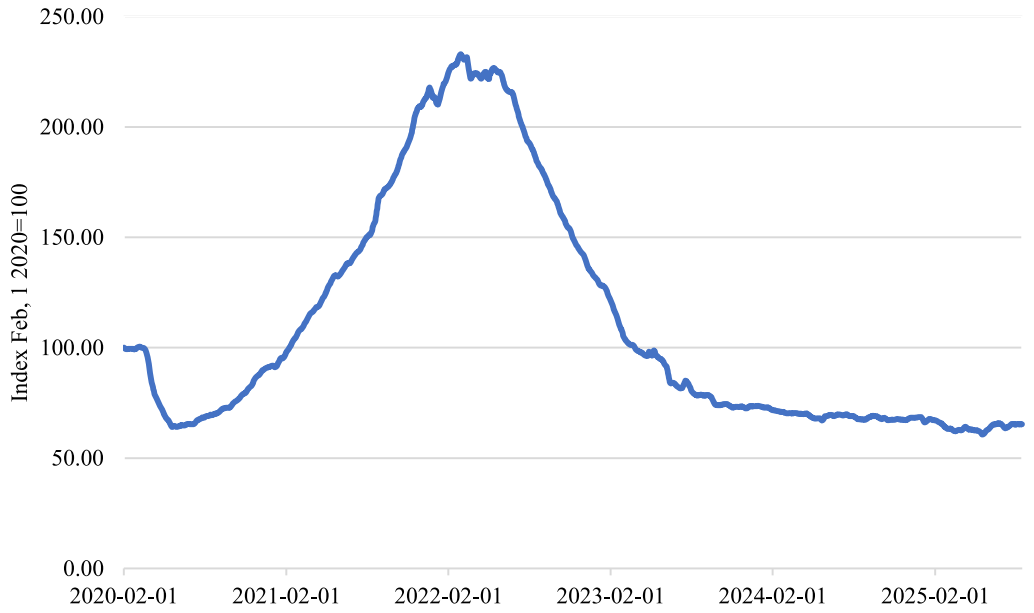



Fig. 6. Software development job postings on indeed in the U.S., 2020–2024. *Source*: adapted from federal reserve economic data [55].

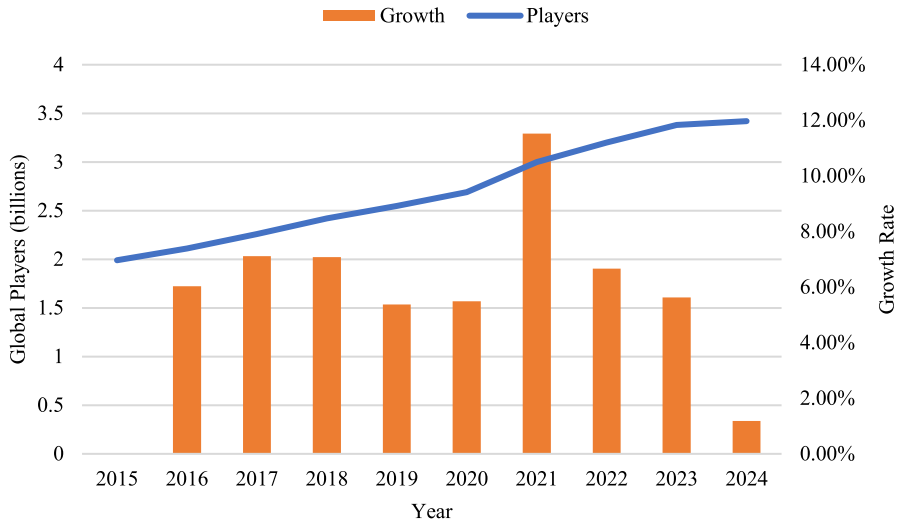



Fig. 7. Global player count and growth rate (2015–2024). *Source*: adapted from Newzoo [63].

pressure from demand-side constraints amid monetary tightening but also aligns with broader industry observations. Based on such data, Newzoo concludes that the global PC and mobile game markets have reached maturity, with limited growth potential in 2025.

## 6 Meso-Level: Industry Dynamics

The wave of M&A among major gaming companies from 2020 to 2022 significantly increased market concentration in the U.S. VGI. This trend was anticipated by Winseck and Yong Jin, who earlier

identified a "oligopolistic core" of large corporations dominating the VGI, surrounded by smaller firms [64]. A decade ago, numerous independent AAA studios—including Obsidian Entertainment, Bungie, Insomniac Games, Respawn Entertainment, and Gearbox Software—remained active in the U.S. Yet, nearly all were acquired during the COVID-19 M&A wave, markedly strengthening oligopolistic tendencies in the U.S. market.

The potential negative impacts of oligopoly on the industry are frequently discussed in literature, such as reduced consumer welfare [65], diminished product quality [66], and suppressed innovation [67]. These trends are already observable in the video game market: despite strong opposition from players, major game companies have decided to raise the standard price of AAA games from $60 in 2020 to $80 in 2025 [68], a 33.3% hike, while the revenue growth of the global VGI during the same period was only 2.5%. Meanwhile, game quality from leading firms showed little improvement, with many flagship products receiving broadly negative reviews from critics and users alike (e.g., References [69–71]).

### 6.1 Rising Costs and Risks

The traditional video game production model carries high risks for investors and developers, as it demands substantial time and capital upfront, while market reception remains uncertain until after release. Investors are exposed to liquidity and market risks [72], while developers face significant tech and cost risks [73].

Meanwhile, modern game development faces significantly increased financial risks due to exponentially rising production costs [74], driven by extended development cycles and team expansion. To meet growing player expectations for performance, narrative, and graphics [75], teams have expanded to include diverse roles such as design, programming, art, sound, management, and quality assurance [76]. However, as developers expand their team size, efficiency and productivity do not maintain a linear positive correlation [77]. This phenomenon, commonly termed Brooks' Law in software project management [78], stems from multiple factors. While initial expansion may benefit from labor specialization [79], exceeding the optimal team size raises management difficulties [80]. The growth in communication channels escalates coordination costs [81], and documentation volume increases disproportionately. Furthermore, necessary sharing of critical assets between departments often triggers interdepartmental conflicts in large corporations [82]. These compounding factors collectively contribute to diseconomies of scale in development [83], ultimately diminishing team productivity and compromising game quality.

Over the decades, the development timelines of video games have been substantially increased (see Figure 8). Release intervals have likewise expanded: Between 2009 and 2015, Naughty Dog consistently put out a new game every two years. Yet after *Uncharted 4* (2015), players waited five years for their next title, *The Last of Us Part II* (2020), and their next game is reported to be released no earlier than 2027. Other U.S.-based AAA studios like Bethesda and Gearbox Software show similar patterns. This temporal expansion, combined with organizational scaling, creates a dual cost multiplier, propelling development costs to unprecedented heights.

For AAA games, marketing expenses that commensurate with development costs significantly increase the total budget. Current AAA games budgets routinely exceed $100 million, with flagship titles like *Call of Duty: Black Ops Cold War* reportedly approaching $700 million in development costs [86]. Since 2020, game development costs in the U.S. have reached the world's highest levels, with per-developer expenditures typically doubling or tripling those in Eastern Europe and China [87]. These soaring costs were manageable during industry expansion but became unsustainable amid revenue stagnation or decline. Public financials reflect the impact: EA's net income declined after 2021 [88], while Take-Two's similarly fell following its acquisition of Zynga in 2022 and turned negative [89]. Escalating costs eroded large corporations' financials while burdening

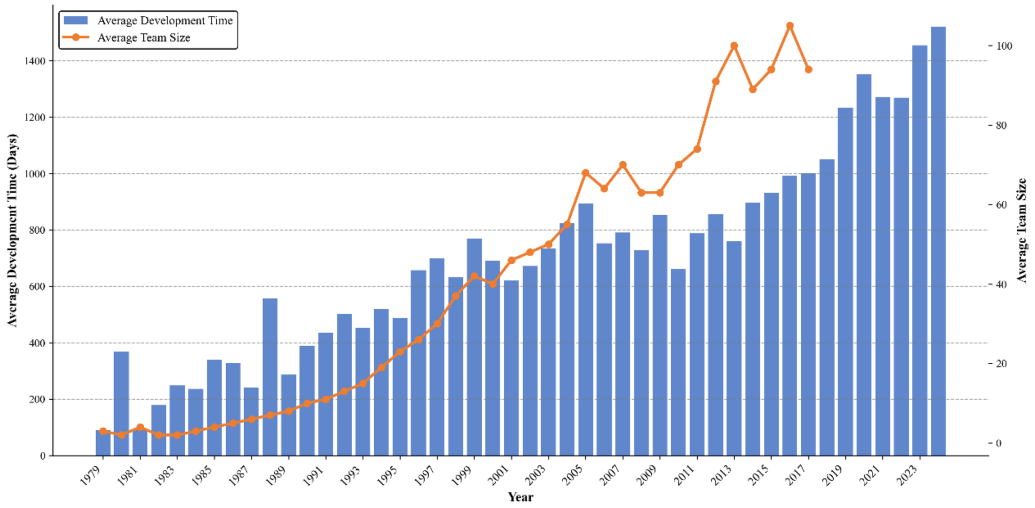



Fig. 8. Average development time and team size for video game projects over time. *Source*: Development time data adapted from SpriteCell [84]; team size data adapted from Mobygames via World Intellectual Property Organization [85]. *Note*: Development time records were manually extracted by SpriteCell from primary sources (developer interviews, official documents, and retrospective publications).

medium enterprises, and U.S. video game companies of all sizes resorted to layoffs to reduce expenses.

### 6.2 Corporate Management Challenges

For major gaming corporations in an industry boom, both investment decision and post-investment management present multifaceted challenges. According to agency cost theory [90], key motives for excessive investment activities in large corporations include empire-building [91], tunneling [92], or even personal preference [93]. Driven by private benefits [94], influential managers in such corporations may invest in **negative net present value** (**NPV**) projects or pet projects [92], both at the expense of shareholder interests. Research also indicates that managerial overconfidence can lead executives to overestimate investment returns, thereby distorting corporate investment decisions [95]. While significantly expanding the size of corporation, these investments and mergers do not necessarily enhance its profitability or enterprise value [96].

Post-M&A corporate integration has also been widely studied, with many studies attributing poor post-acquisition performance to inadequate governance and integration. Overall, factors influencing post-M&A performance are complex, including planning and execution of integration [97], resolution of managerial conflicts of interest [98], establishment of suitable reward systems [99], clarification of corporate vision, and cultivation of a unified organizational culture [100].

Examining M&A cases within the VGI during 2020–22 reveals that many underperformers were directly linked to post-acquisition integration failures (e.g., [101, 102]). For VC-backed small to medium-sized studios, the challenges manifest differently. The highly modular and specialized labor structure required in the game production process compounds these challenges for new ventures [103]. These companies must navigate a range of issues during development, such as controlling cost and quality [104]; managing risk [73]; and handling human capital management [105]. Startups typically lack resources to address modern VGI's specialized complexity, while VCs often lack technical expertise to effectively tackle these challenges. Escalating investment management complexity causes unresolved issues to accumulate, leading to underperforming projects and

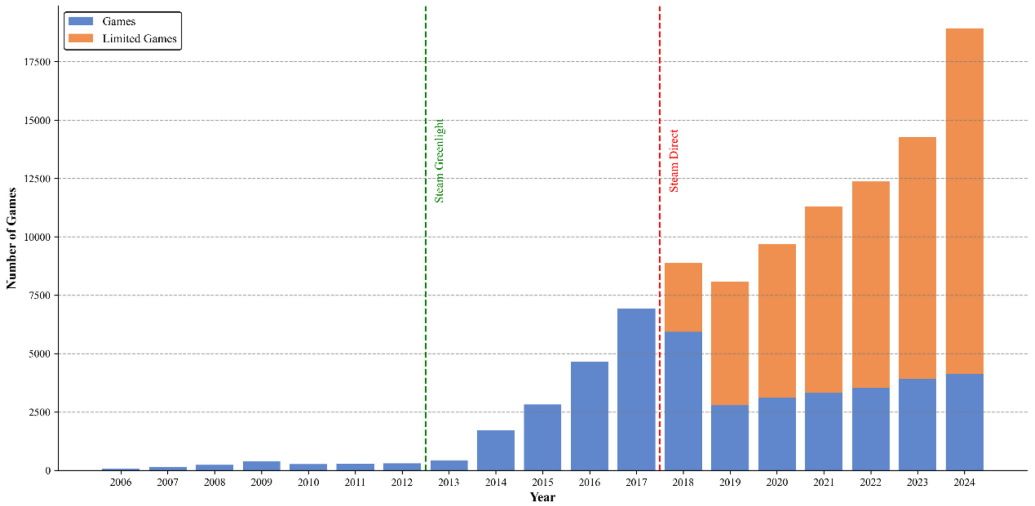



Fig. 9. The number of games released on Steam each year. *Source*: Adapted from SteamDB [38]

subsequent layoffs and studio closures. This phenomenon emerged during the Atari Shock era and has now resurfaced in the post-pandemic VGI after the investment boom.

### 6.3 Hypercompetitive Crowded Markets

The contemporary gaming market faces fierce supply-side competition, particularly evident in platform distribution dynamics. On Steam, the world's largest digital game distribution platform [106], the annual number of new game releases serves as a direct indicator of the intensifying supply-side competition in the VGI.

In 2017, Steam replaced its community-driven Greenlight system with Steam Direct, which allows anyone to publish a game [107]. This transition reduced Valve's oversight of game quality and game supply skyrocketed (see Figure 9). A study revealed that while the number of players on Steam increased linearly, the number of games released grew exponentially [108].

This oversupply has precipitated a pronounced Matthew effect [109], with top-performing titles dominate most of the player base [110]. Third-party statistics further substantiate this phenomenon. According to VGInsights data, 75% of paid games on the Steam platform generate lifetime revenues below $10,000, while the top 1% soar to $7 million [111]. Steam's visibility algorithm, which prioritizes sales performance and wishlist accumulation [112], reinforces market concentration by promoting commercially successful titles. This creates a downward spiral where games which have weak launch performance face algorithmic demotion, further constraining their visibility and recovery potential. In the hyper-competitive environment after the COVID-19 pandemic, the number of games achieving profitability has become even scarcer.

### 6.4 Shifting Player Behaviors

Nowadays, the time players can allocate to gaming is increasingly squeezed by other entertainment forms, such as social media and online video platforms. This phenomenon aligns with attention economy theory [113], which posits that the abundance of information leads to a scarcity of attention. For instance, from 2020 to 2024, the time spent by U.S. adults on online video platforms (e.g., YouTube, Netflix, TikTok) increased by an average of 30% [114] Conversely, time spent playing video games decreased by 26% between 2021 and 2023 [115]. This imbalance between supply

(video games and other entertainment forms) and demand (user attention) underscores the intensifying competition for limited cognitive resources.

Meanwhile, players continue to spend substantial time and funds to established games. Steam's 2024 statistics revealed that only 15% of total playtime was attributed to games released that year [116]. Sales data further confirms this pattern: among Steam's top 12 best-selling products of 2024, only four were new releases, while existing titles like *Counter-Strike 2, DOTA 2, PUBG*, A*pex Legends*, and *Call of Duty* maintained top positions for three consecutive years [117]. This aligns with Newzoo's finding in 2023 that just 66 titles accounted for 80% of all playtime hours across PC and consoles [115], highlighting a strong long-tail effect favoring enduring games.

In the mobile gaming sector, the introduction of Apple's Identifier for Advertisers (IDFA) changes in Q1 2021 has led to the first decline in global mobile game revenue in history [118]. Despite this, players still favor older titles: only one of 2024's top 10 highest-grossing mobile games was released that year [119]. This pattern indicates that, across platforms, players invest time and money predominantly in older games, highlighting a strong lock-in effect [120], making it increasingly challenging for new games to gain traction in a competitive market.

## 7 Micro-Level: Game Development Practices

### 7.1 Live Service Dilemma

In the 1980s, most games were designed for one-time experiences with limited replay value. However, the rise of online games and microtransactions after 2000s has embedded the live service model into players' consuming habits [121]. The 21st century introduced live service titles—such as *World of Warcraft* and *League of Legends* on PC, along with mobile successes like *Honor of Kings* and *Monster Strike*—that have remained popular for over a decade. Through ongoing updates and varied in-game purchase designs, this business model secures stable cash flows for video game companies [122]. By generating reliable revenue that provides resilience against market volatility, the live service approach, also known as **Games-as-a-Service** (**GaaS**) [123], has become vital to the modern video game business.

Although the transition from traditional premium AAA games toward GaaS models began to be explored by video game companies in the 2010s [124], it was not widely adopted due to the high expertise requirements for developers and the need for sophisticated data analytics infrastructures. Leading companies like Sony and Microsoft continued to rely primarily on traditional AAA models in this period. However, in the 2020s, the increasing development cycles and costs of AAA games have inevitably led to gaps in release schedules and unstable cash flow for publishers. Even industry leaders like Microsoft experienced a notable gap, lacking first-party AAA releases for over 12 months since March 2022 [125]. Major corporations begun to perceive GaaS titles as both generators of steady long-term revenue streams and catalysts for strategic transformations, driving substantial acquisitions of such games/developers. For example, under Jim Ryan's leadership, Sony Interactive Entertainment pivoted its strategy toward the GaaS model in 2022 [126]. At its peak, Sony had 12 live service games in development while acquiring studios like Bungie and Firewalk Studios specializing in GaaS development [127]. Similarly, Microsoft bolstered its portfolio through the Activision Blizzard acquisition, gaining long-running live-service franchises including *Call of Duty, World of Warcraft* and *Candy Crush Saga* [45].

While major companies acknowledge GaaS's importance, transitioning from traditional AAA single-player games to a GaaS model remains challenging. This shift requires extensive user metric tracking and analysis [128], with ongoing calibration of in-game economies and monetization based on such data. However, most traditional AAA studios lack these capabilities due to fundamental differences in data paradigms: key live service metrics such as player retention and payer conversion are neither monitored nor relevant in single-player games. Other barriers

include developers' reluctance to adopt online models, inexperience in user operations, data analytics, and economy design, as well as limited agility in integrating feedback via continuous development—all critical for successful GaaS implementation [129]. Moreover, for companies that are new to GaaS models, conservative monetization designs may lead to games failing to break even, while greedy or pay-to-win **microtransaction** (**MTX**) designs may trigger severe player backlash and result in commercial failure [130]. Balancing monetization with player satisfaction requires extensive, ongoing testing and feedback integration. Many initiatives from traditional U.S. AAA and new studios have failed due to a lack of critical GaaS expertise.

### 7.2 Imitative Strategies

The VGI's rising development costs, intensifying competition and escalating risks have led to what scholars term an "innovation dilemma" [131]. Though creativity is crucial for entertainment products, the industry now faces shorter product life cycles, rapidly evolving technologies, and shifting business models. To mitigate the rising risks, developers have grown more conservative and adopted an imitative strategy, focusing on incremental improvements rather than actual innovation [132]. At the same time, numerous studies have shown that imitation can narrow the performance gap between industry leaders and followers [133], or even enable followers to achieve market leadership through imitation [134], further reinforcing this trend in the market. This pattern became particularly evident after 2020, with new AAA projects prioritizing graphical upgrades within established design formulas, rather than exploring new gameplay mechanics.

Another reason for the imitative strategy is that it is well-known the VGI is inherently hit-driven [135]. Few products achieve success, but successful ones generate substantial revenue. The game production process involves high risks, as both developers and publishers find it difficult to predict a product's sales performance and profitability in advance [136]. Accordingly, when a game achieves success, imitators often emerge to replicate its success.

Developers generally agree that there is no clear consensus on the boundary between inspiration and imitation, recognizing it as a complex grey area [137]. Moreover, imitation has become a common practice in the VGI, and some developers acknowledge that imitation is also a part of innovation [138]. However, what developers often overlook is that imitation is subject to diminishing marginal returns [139], and excessive imitation can ultimately exhaust its benefits.

The widely used imitation strategy has led to most original IPs produced by major AAA game companies and released after 2020 being either benchmarked against or imitating existing games. For example, *Starfield* was marketed as "*Skyrim* in space" [140]. However, just like during the early 1980s, as the market becomes saturated with numerous games lacking differentiation, players' patience gradually wanes. The same phenomenon has occurred among gamers since 2022: Due to insufficient differentiation from the games they emulate, media coverage of these imitative titles tends to be blunt, and players' feedback is usually negative (e.g., References [141–143]). Subsequently, most of these games underperformed both critically and commercially.

### 7.3 The Metaverse Hype

The ninth-generation consoles, launched at the end of 2020, introduced advanced graphical technologies like ray tracing [144]. The powerful capabilities of new generation of consoles and the emergence of technologies such as **extending reality** (**XR**) and blockchain seemed to suggest that the previously imagined "metaverse" was on the verge of becoming a reality [145]. Although the concept of the metaverse has never been rigorously defined, due to overly optimistic expectations about the future trends of VGI and a massive influx of capital, the sector began to heavily invest in unproven new technologies. During this period, VCs made significant investments in blockchain-based gaming companies, such as those focused on cryptogaming,

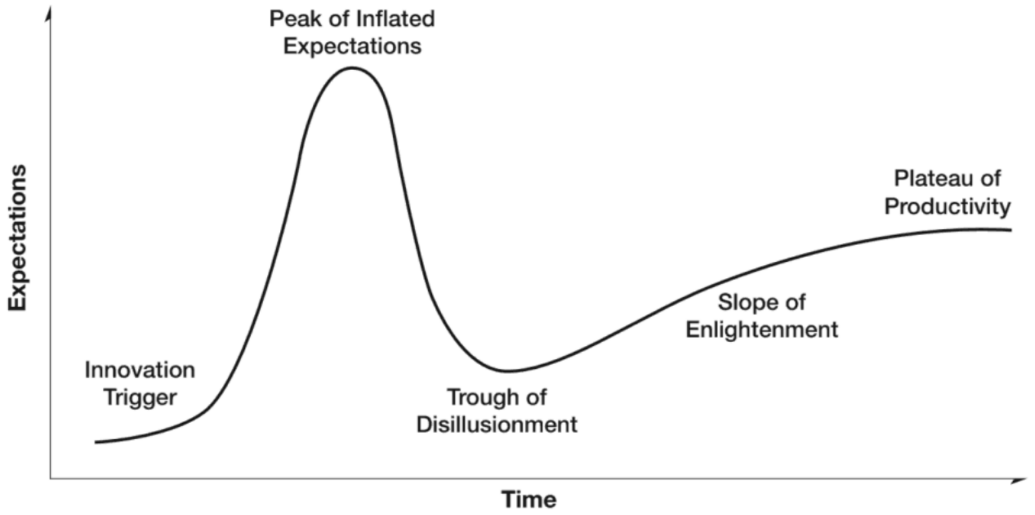



Fig. 10. Gartner's hype cycle. *Source*: Adapted from Fenn and Raskino [154]

**non-fungible tokens** (**NFTs**) and Web3 [146], while major corporations channeled substantial capital into cloud gaming and XR projects [147].

Despite massive funding volumes and transaction activity, Web3 and NFT games have consistently failed to attract mainstream players. Instead, these games remain dominated by market speculators [148]. Due to widespread fraud and rug pulls [149], players exhibit pervasive distrust toward those blockchain games [150]. XR implementations have deficient user experiences [151]. Moreover, compounded by infrastructure limitations and network environment constraints [152], cloud gaming services demonstrate subpar quality-of-experience [153]. Then, most of these metaverse-themed initiatives suffered commercial failures due to a lack of market acceptance.

Gartner's hype cycle model (see Figure 10) is often used to explain the lifecycle of emerging technologies [155]. Recent research indicates that the sudden rise in popularity followed by the rapid cooldown of metaverse-related technologies and concepts aligns with this model [156]. From 2020 to 2022, metaverse-related technologies experienced a period of inflated expectations. However, the once-hot concepts of cloud gaming, XR, and Web3, faded from prominence after 2023 due to lack of market acceptance and the emergence of **artificial intelligence** (**AI**).

The official launch of OpenAI's ChatGPT in late 2022 broke user growth records [157], establishing it as the fastest-growing consumer application, and validating significant market demand for AI tools. Meanwhile, VC pivoted from Web3 investments toward AI, as investments in Web3 games decreased from $ 1.6 billion in Q1 2022 to just $ 221 million by Q1 2024—a drop of over 85% [158]. Concurrently, VC funding for emerging generative AI notably increased from less than $1 billion in Q1 2022 to $ 9 billion in Q4 2024 [159]. The resulting capital drought accelerated Web3 gaming's contraction, culminating in studio layoffs and bankruptcies [160].

Eventually, the job opportunities these technologies had created vanished as funding dried up. Meanwhile, the tech industry's shift toward AI has, to some extent, also exacerbated layoffs [161]. Recent studies have also investigated the AI's unemployment effects: Qin et al. found that AI increases short-term unemployment [162], though the impact varies with time and economic contexts. Wang and Wong project a long-run employment loss of 23% caused by AI [163], with half of these losses occurring within the first five years of transition. Using a triple-difference analysis on 285,000 U.S. firms, Hosseini and Lichtinger demonstrate that generative AI adoption is associated

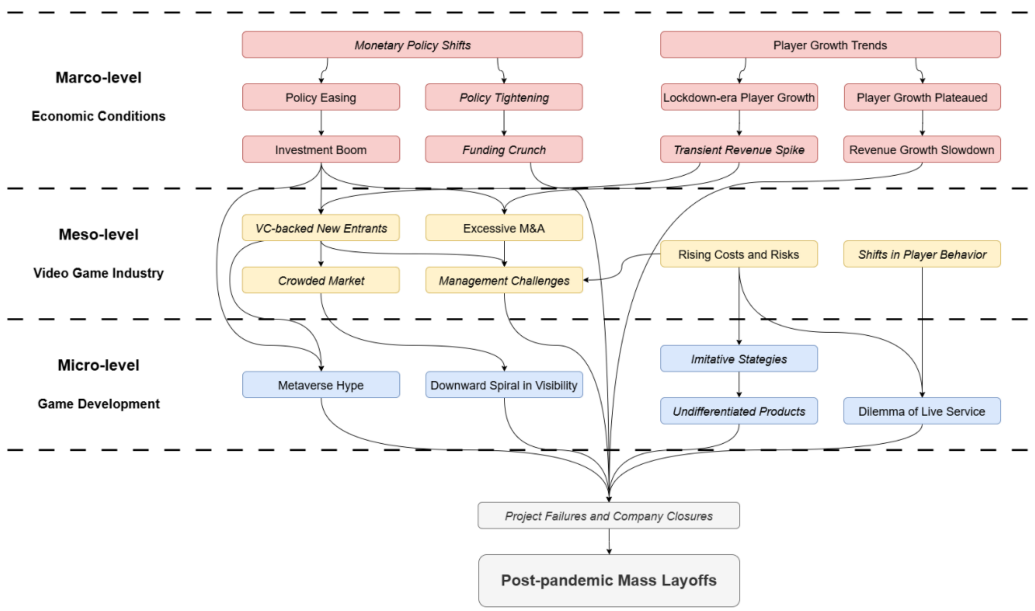



Fig. 11. Summary of factors organized by the three-level framework. *Notes*: Factors in italics were also present during the Atari Shock.

with a sharp decline in junior employment [164]. In short, AI poses a potential exacerbation of unemployment.

## 8 Results: The Outbreak of Mass Layoffs

The causes of this industry-wide layoff crisis emerging after 2022 are diverse and complex, with layoffs at various companies often resulting from multiple concurrent factors from various levels. From a macroeconomic perspective, the sudden rise in interest rate significantly increased the difficulty for video game companies to secure funding as investors retreated from the risky gaming sector (e.g., References [165, 166]), accelerating layoffs among startups relying primarily on equity financing (e.g., References [167–169]). Declining revenue and stagnant player growth, alongside pandemic-driven investment boom, led to market saturation and fierce competition. To mitigate risks from rising costs and fierce competition, major gaming corporations adopted conservative and imitative strategies. While established intellectual property (IP)-backed products leveraged existing fanbases, imitative products without an IP faced significantly greater challenges. Players grew increasingly weary of plentiful clones, causing many such games to fail commercially and leading the developers to implement mass layoffs or studio closures (e.g., References [170–172]). Meanwhile, AAA studios attempting to pivot from traditional single-player models to GaaS have consistently failed due to a lack of core live service skills: user analytics, in-game event management, and virtual economy design (e.g., References [173–175]).

Furthermore, excessive mergers and acquisitions created significant management challenges, including frequent shifts in parent company strategy and ineffective control of subsidiaries, which also contributed to numerous project failures (e.g., References [170, 176, 177]). VC-backed small and medium-sized studios demonstrated weaker risk resilience due to limited development capability and expertise, usually releasing low-quality games that met commercial failure (e.g., References [178–180]). Simultaneously, many metaverse-focused video game companies or internal divisions of major corporations, born of abundant investments during COVID, faced substantial layoffs and

closures due to poor market acceptance (e.g., References [181–183]). Finally, AI emerged as the latest accelerant of this layoff wave, as widespread adoption negatively impacted programming and art positions, with reports ironically suggesting game developers being replaced by the very AI tools they created [184].

Ultimately, this triggered a supply-side layoff crisis across the global VGI: Job losses surpassed 10,500 in 2023, surged to 14,600 in 2024, and added 4,000 more by July 2025—with the U.S. consistently suffering the deepest cuts throughout 2023–2025 [3]. In U.S., where the at-will employment doctrine prevails and dismissal protections are weak [185], major publicly listed gaming companies implemented large-scale workforce reductions (e.g., References [186–188]), with some cutting thousands of positions, alongside the cancellation of numerous game projects (e.g., References [171, 189, 190]). Similarly, numerous multinational gaming corporations have implemented either job cuts or full closures of their U.S.-based development studios (e.g., References [172, 191, 192]). Concurrently, layoffs and shutdowns among small-to-medium-sized U.S. video game companies became increasingly frequent (e.g., References [193–195]).

As of 2025, while the pace of layoffs has slowed, the trend persists. Meanwhile, investment activities in the VGI have seen a decline in frequency but have instead shifted toward larger transaction sizes. For instance, in September 2025, Electronic Arts was privatized for $55 billion, marking the largest PE leveraged buyout (LBO) in history [196]. However, analysts and employees have expressed widespread concerns about potential layoffs following LBOs [197]. Given that game development cycles have now generally extended to around five years, projects initiated during the 2020–2022 investment boom may still require an additional 1–2 years to be fully digested. From this perspective, the U.S. VGI, which has been most heavily impacted by the current wave of layoffs, may still experience a period of layoffs before recovery materializes.

## 9 Discussion: A Historical Comparison

Through the preceding analysis, it is clear that, due to differing historical contexts, several contemporary factors driving the current wave of layoffs in the VGI—such as soaring development costs, the exhaustion of the player demographic dividend, management challenges from overexpansion, failures in live service model transitions, and the collapse of metaverse concepts due to market rejection—were absent during the Atari Shock era.

However, several factors parallel the Atari Shock era (see Figure 9), including a VC-fueled influx of new entrants, monetary tightening, market saturation, fierce competition, changing player preferences, and widespread game cloning. Within the "3Ds" framework of the Atari Shock – *Declining entry barriers, Decreased product differentiation*, and *Disruptive innovation* – at least two and a half mechanisms recur today. First, lower platform barriers and excessive COVID-era investments led to market saturation, reflecting *Declining barriers*. Second, imitative products developed by AAA studios mirrors past *Decreased differentiation*. Third, while no true *Disruptive innovation* has threatened gaming, player attention shifting to social media and video platforms partly echo the migration to emerging platforms (such as PCs) during the Atari Shock. Beyond the “3Ds”, monetary tightening remains a critical yet overlooked precursor in both crises [198].

Notably, the overall impact on the VGI appears less severe than during the Atari era. Unlike the Atari Shock, which devastated consumer demand, the current layoffs have primarily affected the industry’s supply side while leaving demand relatively stable. This resilience stems from two major factors: globalization and diversification.

### 9.1 Globalization

Historically, the global VGI expanded from its initial dominance by North America and Japan to include Western Europe, followed by South Korea, Eastern Europe, and eventually China, which

leveraged its vast population and mobile internet uptake to become the largest market. This growth was driven by broader global capitalism and industry globalization in the early 21st century [199]. Market instability in the West also drove firms to reduce costs through structural and geographic expansion [32]. These factors accelerated the globalization of production networks and spurred an explosion in the worldwide player base, transforming gaming into a global phenomenon.

Unlike the early 1980s when players could only buy games from local developers and publishers at physical stores, online distribution platforms now make games from around the world accessible to every player [200]. This worldwide accessibility guarantees that even when production issues, like oversupply or deteriorating quality, impact one region, gamers can still buy games developed in regions that are less affected. Thus, while the current layoff crisis has hit North America hardest [4], Europe experienced a less severe investment bubble during the 2020–2022 boom, with countries like Germany less affected by production issues due to fewer mergers and stronger dismissal protections [201]. Similarly, Japan, where mergers were minimal and which maintains a tradition of lifetime employment [202], also experienced relatively limited layoffs. In reality, games from Europe and Japan, such as *Baldur's Gate 3, Helldivers 2, The Legend of Zelda: Tears of the Kingdom*, and *Palworld*, have achieved commercial success in the video game market after 2023.

### 9.2 Diversification

The diversification of gaming platforms and business models has significantly bolstered the industry's ability to withstand disruptions. Unlike the Atari era, when players were confined to arcade and home console games, today's gamers have a variety of options across mobile devices, PCs, and consoles [203]. Mobile platforms offer a wide range of experiences, from full-fledged apps to instant-play mini-games [204]. This variety ensures that even if game production and quality one segment of the industry are negatively impacted, players can easily shift their attention to other platforms.

As previously discussed, the live service model has also provided the VGI with stable cash flows and now constitutes a significant portion of industry revenue. This represents another form of diversification in revenue channels. Moreover, it is noteworthy that regions like China and Japan, which have a strong presence of mobile and live service game developers, have been less impacted by the recent wave of layoffs compared with regions like North America and Europe, which are dominated by the traditional premium business model. This resilience can be partly attributed to the advantages of diversification in releasing platforms and business models, which mitigates risks and ensures stability amid industry fluctuations.

## 10 Conclusion

The mass layoff crisis that has gripped the U.S. VGI since 2023 is a multifaceted phenomenon rooted in interconnected economic shifts, industry-specific vulnerabilities, and challenges in game development. This crisis, while distinct in its contemporary context, echoes the structural fragilities of the Atari Shock in 1983, particularly in its exposure to market oversaturation, product homogeneity, and the shifts of consumer habits. However, the modern industry's resilience—shaped by globalization and platform diversification—has prevented a collapse of the scale witnessed in the 1980s.

At its core, the 2023–2025 crisis emerged from a confluence of factors: macroeconomic tightening—marked by rising interest rates and liquidity contraction—that reversed the pandemic-era investment boom; industry-wide pressures, including inflated development costs and managerial inefficiencies; and game production challenges, such as failed transitions to live-service models, overreliance on imitative design, and the collapse of metaverse-related projects due to market rejection. Collectively, these factors triggered a supply-side crisis, with layoffs cascading across large corporations, mid-sized studios, and VC-backed startups alike.

Comparisons with the Atari Shock reveal parallels: both crises were characterized by crowded market caused by the rising number of new entrants, imitative strategy that leads to homogenized products, and shifts in player behaviors. Yet critical differences define the modern landscape. Unlike the Atari era's near-total demand destruction, today's industry demonstrates remarkable shock absorption through two adaptations: First, globalized production networks mitigate regional instability, as evidenced by limited layoffs in regions with robust employment protections. Second, diversified revenue streams—particularly mobile gaming and live service models—create counter-cyclical stability that 1980s console-dependent markets lacked. These structural buffers transformed the crisis into a supply-side correction rather than systemic collapse.

For the VGI, sustainable recovery requires the implementation of multifaceted strategies. From a game development perspective, studios should prioritize transcending imitation-driven development by focusing on differentiation. Developers can also consider controlling team expansion speed and development cycle length, which can effectively reduce management complexity for both companies and projects—as demonstrated by recent successes like *Clair Obscur: Expedition 33* [205].

In sum, the current crisis serves as a critical reminder that the VGI's long-term sustainability depends on learning from history while adapting to new realities. By internalizing historical parallels while leveraging contemporary advantages, I hope that the VGI can transform vulnerability into resilience.

## Data Availability Statement

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

## Declaration of Conflicting Interests

The author is employed by a video game company. However, this study was conducted as an independent academic project without organizational involvement. All data are sourced from publicly available repositories cited in references. The author confirms no financial interests or personal relationships that could influence the research outcomes.

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